

Threshold Pairing

Time Limit: 1.0 seconds

Memory Limit: 512 MB

Chef is organizing a mega inventory sale. The warehouse contains N items, where the i -th item has a value denoted by A_i . Items with the same value belong to the same category.

To qualify for a special wholesale bundle deal, Chef must select exactly two distinct item categories, say category x and category y ($x \neq y$), such that:

- At least one of the selected categories appears at least P times in the inventory.
- The other selected category appears at least Q times in the inventory.

When a category x with frequency f_x is selected, all f_x items of that category are bundled together, yielding a total value equal to $x \times f_x$.

Chef wants to maximize the total combined value of the two selected categories, which is:

$$\text{Total Value} = (x \times f_x) + (y \times f_y)$$

Help Chef find the maximum possible total value he can obtain. If it is impossible to pick two distinct categories satisfying the requirements, output 0.

Input

The first line contains a single integer T ($1 \leq T \leq 10^4$) — the number of test cases.

For each test case:

- The first line contains three integers N, P, Q ($2 \leq N \leq 2 \times 10^5$, $1 \leq P, Q \leq N$) — the number of items in the inventory and the two threshold parameters.
- The second line contains N space-separated integers $A_1, A_2, \dots, A_n$ ($1 \leq A_i \leq 10^9$) — the values of the items.

It is guaranteed that the sum of N over all test cases does not exceed 2×10^5 .

Output

For each test case, print a single integer representing the maximum total value obtainable from two distinct categories satisfying the conditions. Print 0 if no valid pair of categories exists.

Sample Input

```
3
7 2 3
1 1 1 2 2 3 3
6 2 2
5 5 5 10 10 10
4 3 3
1 1 2 2
```

Sample Output

9
45
0

Explanation of Samples

- Test Case 1:
 - Category frequencies:
 - Value 1: frequency = 3, total category value = $1 \times 3 = 3$
 - Value 2: frequency = 2, total category value = $2 \times 2 = 4$
 - Value 3: frequency = 2, total category value = $3 \times 2 = 6$
 - The required thresholds are $P = 2$ and $Q = 3$ (one category must have frequency ≥ 3 , the other ≥ 2).
 - Category 1 qualifies for the threshold 3.
 - Categories 2 and 3 qualify for the threshold 2.
 - Best pair is category 1 and category 3: total value = $3 + 6 = 9$.
- Test Case 2:
 - Categories 5 and 10 both appear 3 times (which is ≥ 2).
 - Total value = $(5 \times 3) + (10 \times 3) = 15 + 30 = 45$.
- Test Case 3:
 - Both P and Q require at least 3 occurrences. No category appears 3 times, so no valid pair can be formed. Output is 0.